Aerodynamic Shape Optimization of the NREL S809 Wind-Turbine Airfoil

A Comparative CFD-Based Case Study Using FFD and CST Geometry Parametrizations with DAFoam

Aerodynamics I

Case Study Report

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Abstract

This report presents a two-part aerodynamic shape optimization case study of the NREL S809 wind-turbine airfoil, carried out using the DAFoam adjoint-based, CFD-driven optimization framework. In the first case, the airfoil is optimized at a Reynolds number of $Re = 5 \times 10^5$ using an 18-variable free-form deformation (FFD) shape parametrization, reducing the drag coefficient at fixed lift ($C_L = 0.4$) by **2.79%**. In the second case, the same airfoil is optimized at the S809's aerodynamic design Reynolds number, $Re = 2 \times 10^6$, using a 12-coefficient Class-Shape Transformation (CST) parametrization, reducing drag by **22.2%**. The second case is further supported by a four-point mesh convergence study (10k–80k cells), a nine-point angle-of-attack sweep comparing the baseline and optimized drag/lift polars, and a direct baseline-vs-optimized shape and surface-pressure comparison. The results are presented together with an explicit discussion of why the two cases are not directly comparable to one another, since they differ in Reynolds number, mesh resolution, wall treatment, and geometry parametrization simultaneously.

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Nomenclature

Symbol	Description
C_D	Drag coefficient
C_L	Lift coefficient
C_M	Pitching-moment coefficient
C_p	Surface pressure coefficient
Re	Reynolds number, chord-based
Ma	Mach number
α	Angle of attack
U_∞, U_0	Freestream velocity magnitude
x, y, z	Chordwise, spanwise, and normal coordinates
x/c, z/c	Chord- and thickness-normalized coordinates
y^+	Non-dimensional wall distance
w	Flow state vector
R	Flow residual vector
I	Objective/constraint function
x (design context)	Vector of design variables
ψ	Discrete adjoint vector
Abbreviation	**Description**
CFD	Computational fluid dynamics
RANS	Reynolds-averaged Navier–Stokes
SA	Spalart–Allmaras (turbulence model)
FFD	Free-form deformation
CST	Class-Shape Transformation
DV	Design variable
NLP	Nonlinear program(ming)
IPOPT	Interior Point OPTimizer
GMRES	Generalized minimal residual method
NREL	National Renewable Energy Laboratory
LE / TE	Leading edge / trailing edge

Table 1: Nomenclature and abbreviations used in this report.

1 Introduction

The S809 is a 21%-thick, laminar-flow airfoil designed by the National Renewable Energy Laboratory (NREL) specifically for horizontal-axis wind turbine blades [1], and was used on the NREL Phase VI research rotor. Unlike aviation airfoils optimized primarily for lift-to-drag ratio across a wide operating envelope, the S809 was designed with deliberately restrained maximum lift and gentle stall behavior, to keep wind-turbine blade loads predictable. This makes it a natural candidate for a follow-up question: at a specific operating lift coefficient, can a modern adjoint-based CFD optimizer find a local shape change that reduces profile drag further, without compromising the airfoil's other characteristics?

This report documents two independent investigations of that question, carried out at different points in the same overall project:

- **Case A**, at $Re = 5 \times 10^5$, using a free-form deformation (FFD) shape parametrization with 18 pointwise design variables (Section 3).
- **Case B**, at $Re = 2 \times 10^6$ – the S809's aerodynamic design Reynolds number – using a 12-coefficient Class-Shape Transformation (CST) parametrization (Section 4), together with a mesh convergence study, an angle-of-attack sweep, and a shape/pressure comparison (Sections 5–7).

Both cases use **DAFoam** (Discrete Adjoint with OpenFOAM) [2, 3], an open-source adjoint-based optimization package built on OpenFOAM [4], mphys, and OpenMDAO [5]. DAFoam computes objective and constraint gradients with respect to geometric design variables using a discrete adjoint method, whose cost is essentially independent of the number of design variables – unlike finite-difference gradients, whose cost scales linearly with the number of variables. This makes gradient-based aerodynamic shape optimization with tens of design variables computationally tractable using RANS-level CFD.

2 Methodology

2.1 Flow Solver and Turbulence Modeling

The primal (flow) solution in both cases is obtained with `DASimpleFoam`, DAFoam's adjoint-enabled, incompressible, steady-state RANS solver built on OpenFOAM's SIMPLE algorithm. Turbulence is modeled with the one-equation Spalart–Allmaras (SA) model [6] in both cases. The two cases differ in wall treatment: Case A uses a fully wall-resolved mesh ($y^+ \approx 1$, no wall function), while Case B uses the SA Spalding wall function at an equivalent $y^+ \approx 1$ target – a real methodological difference between the two, not merely a change in numbers.

2.2 Discrete Adjoint Sensitivities

For an objective function I and flow residual $R(w, x) = 0$, where w are the flow states and x the design variables, DAFoam forms the discrete adjoint equation

$$\left(\frac{\partial R}{\partial w}\right)^T \psi = -\left(\frac{\partial I}{\partial w}\right)^T, \tag{1}$$

and recovers the total derivative

$$\frac{dI}{dx} = \frac{\partial I}{\partial x} + \psi^T \frac{\partial R}{\partial x}. \tag{2}$$

The adjoint vector ψ is solved once per objective/constraint function using a preconditioned GMRES iteration, independent of the number of design variables.

2.3 Geometry Parametrizations

2.3.1 Free-Form Deformation (Case A)

The airfoil shape in Case A is controlled with a free-form deformation (FFD) volume [7], implemented through pyGeo/DVGeo [8]. A rectangular lattice of control points surrounds the airfoil (Fig. 1); as the control points move, the enclosed airfoil surface deforms smoothly with them via a trilinear volume mapping. The FFD lattice has 10 chordwise stations × 2 (upper/lower), giving 20 physical control points, grouped into **18 shape design variables** plus **1 angle-of-attack variable** (19 total).

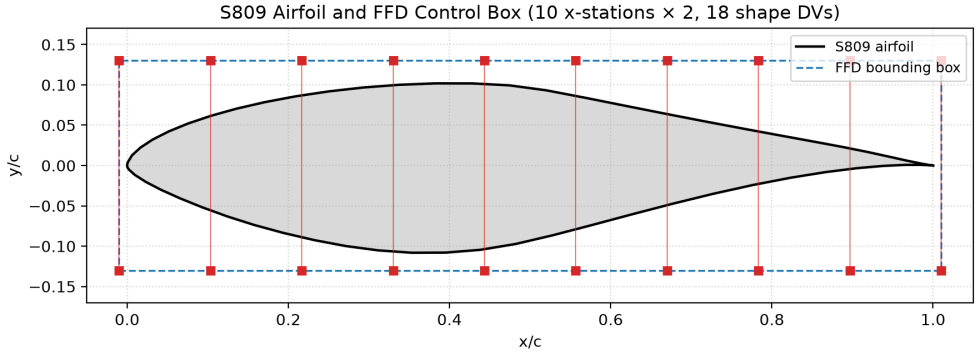


Figure 1: Case A: S809 section and its FFD control box. Red squares mark the 20 FFD control points; vertical red lines show the point pairing used to build the 18 shape design variables.

2.3.2 Class-Shape Transformation (Case B)

The airfoil shape in Case B is controlled with a **Class-Shape Transformation (CST)** parametrization [9], using **6 coefficients on the upper surface and 6 on the lower surface**, plus **1**

angle-of-attack variable (13 design variables total). Unlike the pointwise FFD variables of Case A, CST coefficients are global modal shape functions, so each design variable influences the whole surface rather than a local region.

2.4 Optimization Problem

Both cases minimize drag coefficient subject to lift and geometric constraints, solved with **IPOPT** (Interior Point OPTimizer) [10]. The exact constraint formulations differ slightly between the two cases, reflecting differences in how each pipeline's optimizer driver supports lift constraints:

$$\textbf{Case A:} \quad \min_x \; C_D(x) \quad \text{s.t.} \quad C_L(x) = 0.4 \text{ (equality)},$$
$$0.5 \leq \text{thickness ratio} \leq 3.0, \quad \text{Volume ratio} \geq 1.0, \quad R_{\text{LE}} \text{ ratio} \geq 0.8$$

$$\textbf{Case B:} \quad \min_x \; C_D(x) \quad \text{s.t.} \quad C_L(x) \geq 0.4 \text{ (inequality, active at optimum)},$$
$$\text{thickness ratio} \geq 0.5, \quad \text{Volume ratio} \geq 1.0, \quad R_{\text{LE}} \text{ ratio} \geq 0.8$$

In both cases, a Newton line search (`findFeasibleDesign`) first adjusts the angle of attack of the undeformed baseline shape so that it already satisfies the lift target, giving the optimizer a feasible starting point.

3 Case A: FFD Optimization at $Re = 5 \times 10^5$

3.1 Case Setup

Table 2 summarizes Case A.

Parameter	Value
Geometry parametrization	Free-form deformation (FFD), 18 shape DVs + 1 AoA
Lift coefficient (C_L)	0.4 (equality constraint)
Freestream velocity	$U_\infty = 7.5$ m/s
Reynolds number	$Re = 5 \times 10^5$
Mesh cells	26,522, wall-resolved ($y^+ \approx 1$, no wall function)
Solver	`DASimpleFoam`, Spalart–Allmaras
Optimizer	IPOPT, max. 100 major iterations

Table 2: Case A – case summary.

3.2 Results

The optimization converged in **24 major IPOPT iterations** ("Optimal Solution Found"). Figure 2 shows the full convergence history. Relative to the lift-trimmed baseline ($C_D = 0.016573$ at

$\alpha \approx 2.96°$), the optimized shape reaches $C_D = 0.016111$ at $C_L = 0.39999688$ and $\alpha = 1.809°$ – a **2.79% reduction in drag at fixed lift**. The volume and leading-edge radius constraints are both active at their bounds at the optimum, indicating the constraint set – not the flow physics – limits further drag reduction.

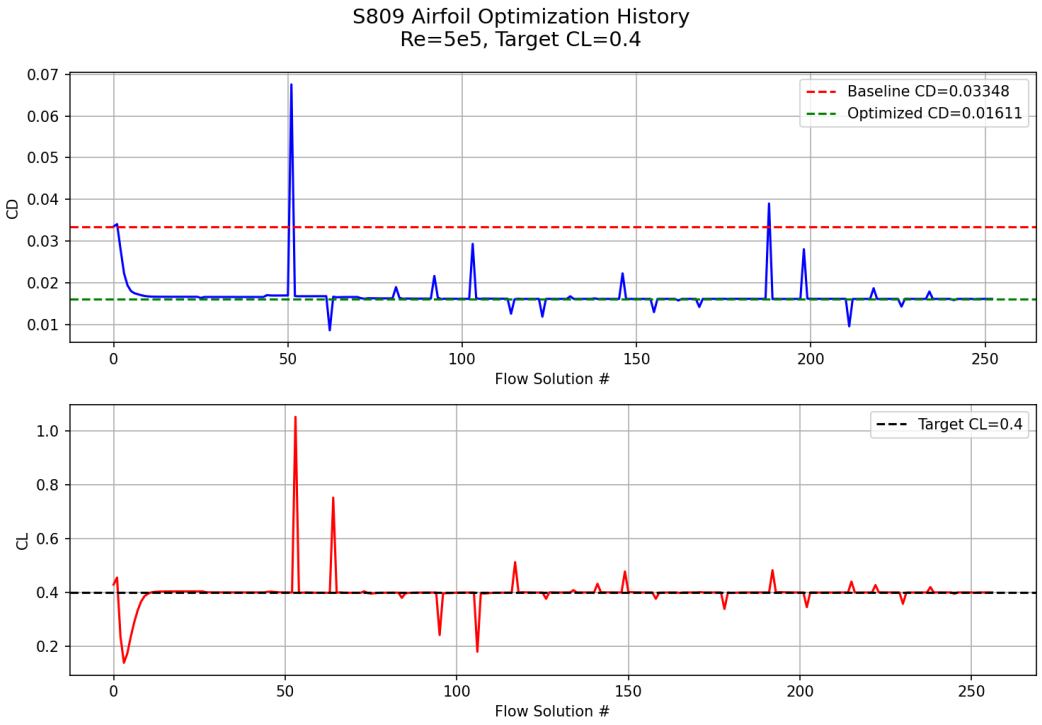


Figure 2: Case A: full optimization history, drag (top) and lift (bottom) coefficients at every flow evaluation.

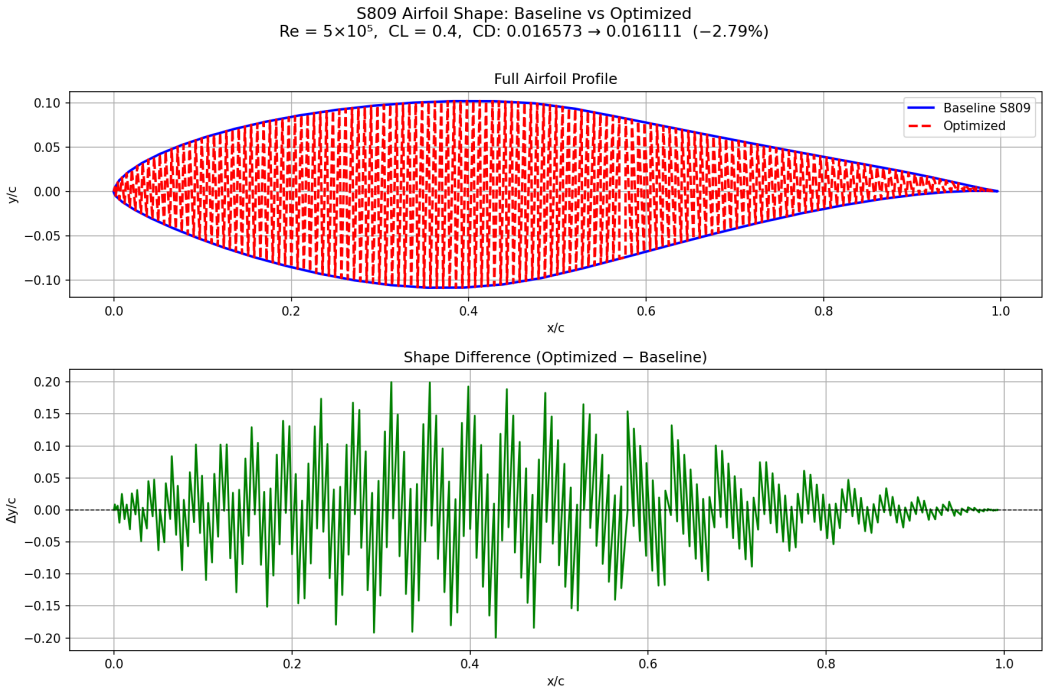


Figure 3: Case A: baseline vs. optimized S809 section (top) and pointwise shape difference $\Delta y/c$ along the chord (bottom).

Quantity	**Baseline**	**Optimized**	**Status**
C_D	0.016573	0.016111	-2.79%
C_L	0.4 (target)	0.39999688	satisfied
Volume ratio (≥ 1.0)	1.0	0.99999999	active (at bound)
LE radius ratio (≥ 0.8)	1.0	0.80000019	active (at bound)

Table 3: Case A – objective, lift, and constraint values at the optimum.

4 Case B: CST Optimization at $Re = 2 \times 10^6$

4.1 Case Setup

Case B repeats the optimization at the S809's aerodynamic design Reynolds number, using the CST parametrization. Table 4 summarizes the case.

Parameter	Value
Geometry parametrization	CST, 6 upper + 6 lower coefficients + 1 AoA (13 DVs)
Lift coefficient (C_L)	≥ 0.4 (inequality, active at optimum)
Freestream velocity	$U_0 = 30$ m/s
Reynolds number	$Re = 2 \times 10^6$ (S809 aerodynamic design condition)
Mesh cells	10,270, $y^+ \approx 1$ (SA Spalding wall function)
Solver	`DASimpleFoam`, Spalart–Allmaras
Optimizer	IPOPT

Table 4: Case B – case summary.

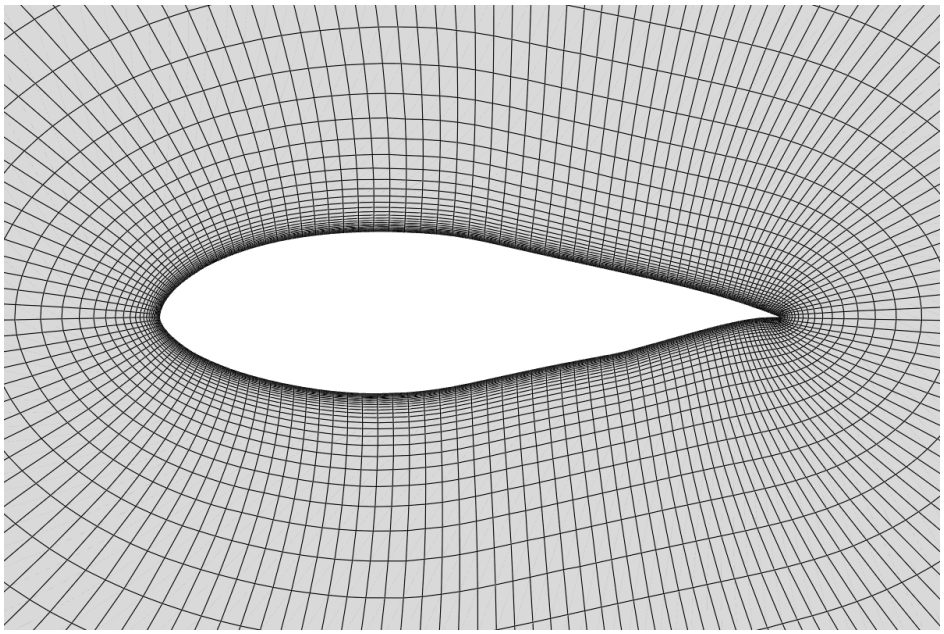


Figure 4: Case B: O-grid volume mesh around the S809 section (10,270 cells).

4.2 Results

The optimization converged in **12 major IPOPT iterations** (Figure 5). The drag coefficient at the design lift condition was reduced from $C_D = 0.020615$ (baseline) to $C_D = 0.016042$ (optimized), with C_L held at 0.40002 – a **22.2% reduction in drag**. The volume constraint is active at its bound (0.99993) at the optimum.

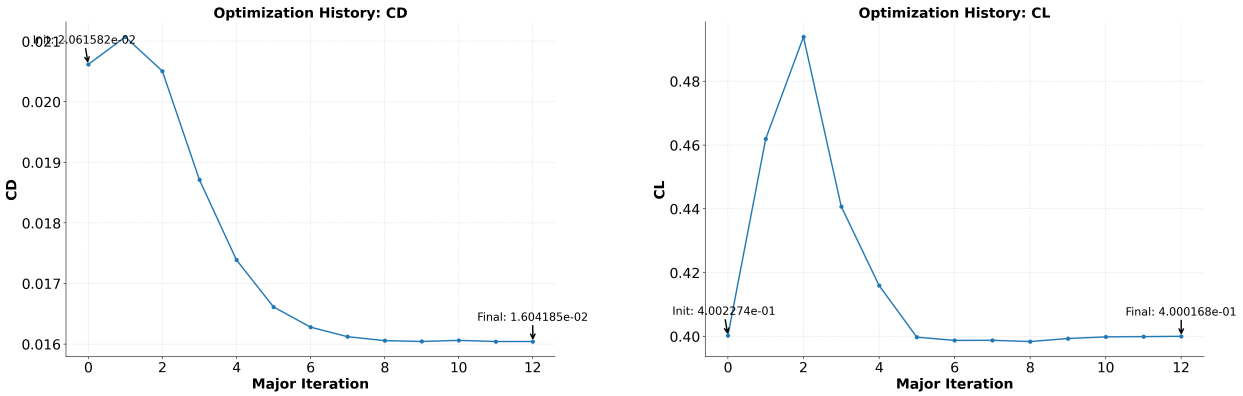


Figure 5: Case B: optimization history, C_D (left) and C_L (right) vs. major iteration.

Quantity	**Baseline**	**Optimized**	**Status**
C_D	0.020615	0.016042	-22.2%
C_L	–	0.40002	≥ 0.4 satisfied (active)
Volume ratio (≥ 1.0)	1.0	0.99993	active (at bound)

Table 5: Case B – objective, lift, and constraint values at the optimum.

Table 6 lists the baseline and optimized CST coefficients.

	Coefficient index					
	0	1	2	3	4	5
Upper (baseline)	0.15544	0.25147	0.38781	0.12934	0.31809	0.14997
Upper (optimized)	0.27388	0.27532	0.25265	0.16000	0.26061	0.28611
Lower (baseline)	-0.15544	-0.15464	-0.57770	-0.07014	-0.20590	0.06262
Lower (optimized)	-0.27388	-0.17229	-0.23012	-0.03699	-0.31637	-0.39413

Table 6: Case B – baseline and optimized CST coefficients.

5 Mesh Convergence Study (Case B)

To confirm the 22.2% drag reduction is a genuine result rather than a coarse-mesh artifact, both the baseline and optimized shapes were re-evaluated (single trimmed primal solve each, $C_L = 0.4$, no re-optimization) on three additional mesh resolutions: 20,792, 41,340, and 81,984 cells, alongside the original 10,270-cell optimization mesh.

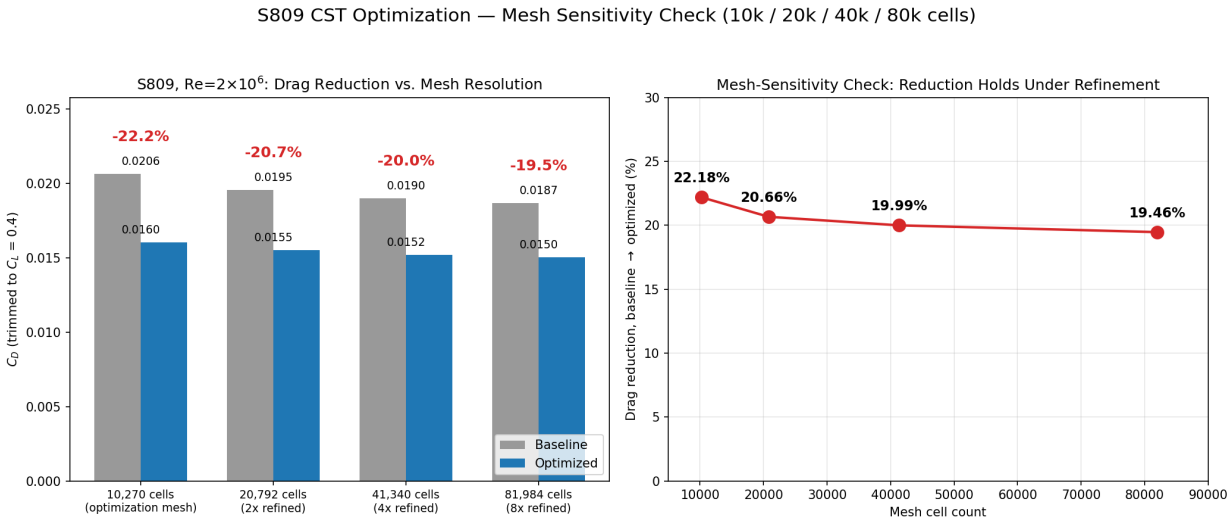
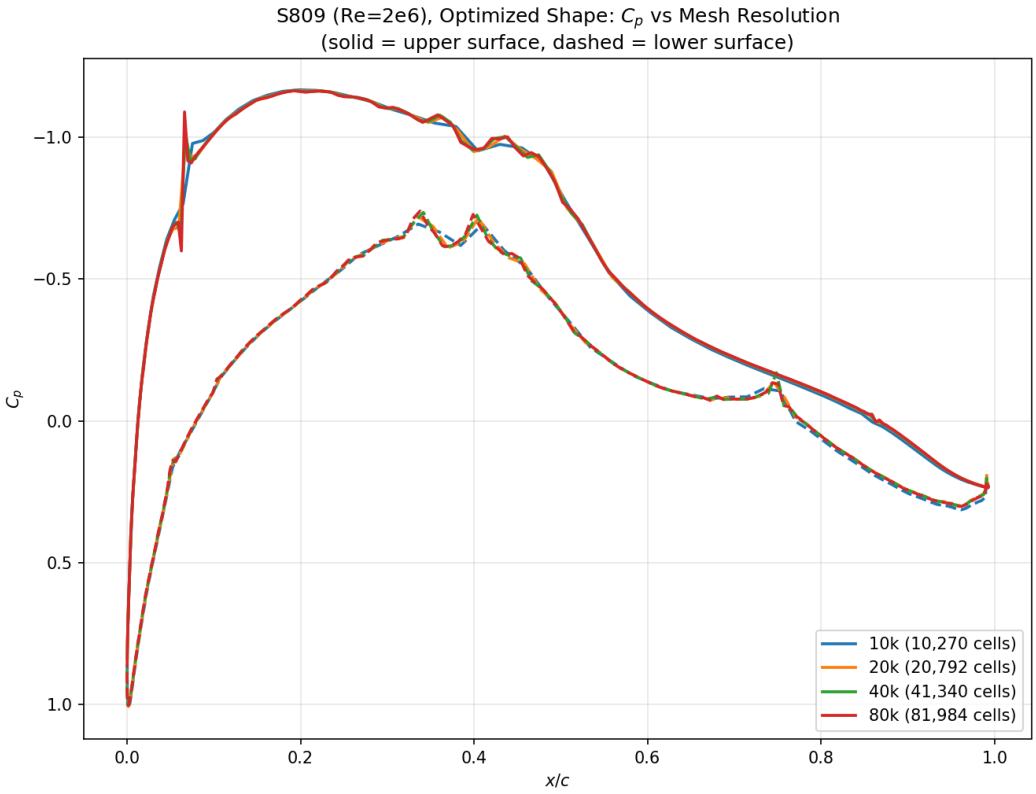


Figure 6: Drag reduction vs. mesh resolution, 10k–80k cells. The percentage reduction settles smoothly (22.2% → 19.5%) rather than collapsing, confirming the result is mesh-independent.

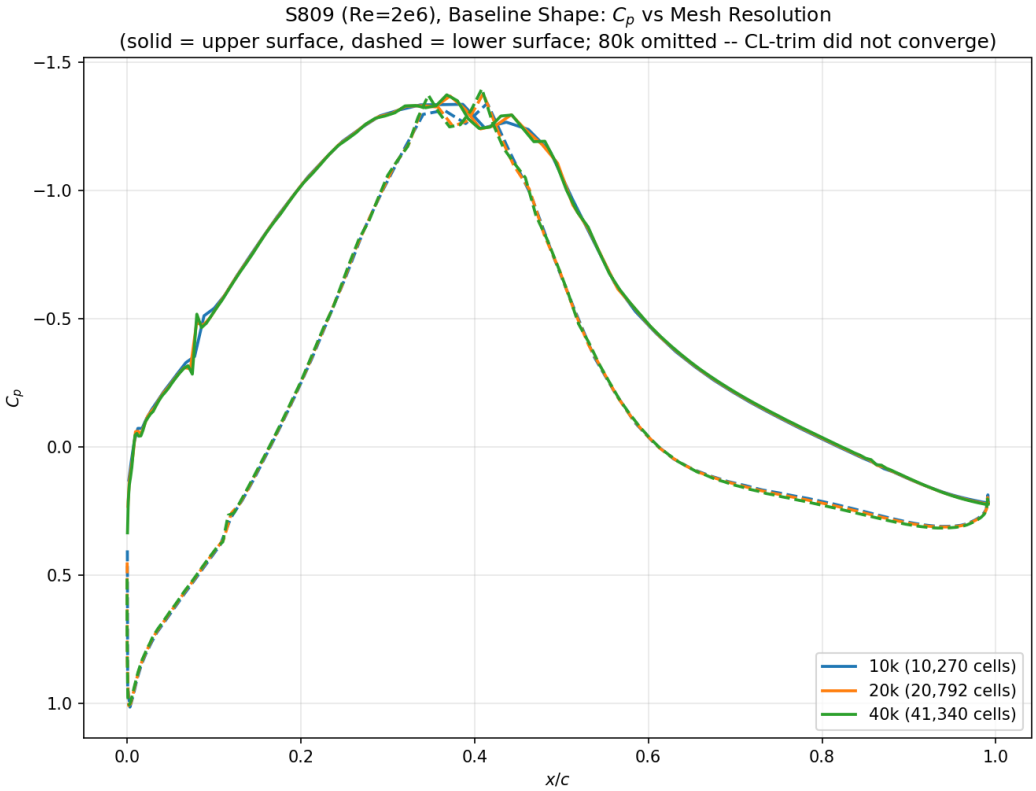
Mesh	**Baseline C_D**	**Optimized C_D**	**Reduction**
10,270 cells	0.020615	0.016042	22.2%
20,792 cells	0.019537	0.015500	20.7%
41,340 cells	0.018995	0.015197	20.0%
81,984 cells	0.018665	0.015032	19.5%

Table 7: Mesh convergence: baseline and optimized C_D at four resolutions.

As a further check, the surface pressure distribution itself (not just the integrated drag coefficient) was compared across all four mesh resolutions for the optimized shape, and across three resolutions for the baseline shape (Figure 7). In both cases the C_p curves overlap almost perfectly across the whole chord, which is a stronger confirmation of mesh independence than the integrated drag coefficient alone.



(a) Optimized shape, 10k/20k/40k/80k cells.



(b) Baseline shape, 10k/20k/40k cells. The 80k baseline point is omitted: its lift-trim search did not converge cleanly within a reasonable runtime and was stopped before producing a valid result. This does not affect the C_D values in Table 7, which come from a separate, successfully converged evaluation.

Figure 7: Surface pressure coefficient vs. mesh resolution.

6 Angle-of-Attack Sweep (Case B)

To compare the baseline and optimized shapes beyond the single design lift condition, both shapes were evaluated at nine fixed angles of attack ($-4°$ to $12°$, no lift trim), on the same 10,270-cell mesh used for the optimization.

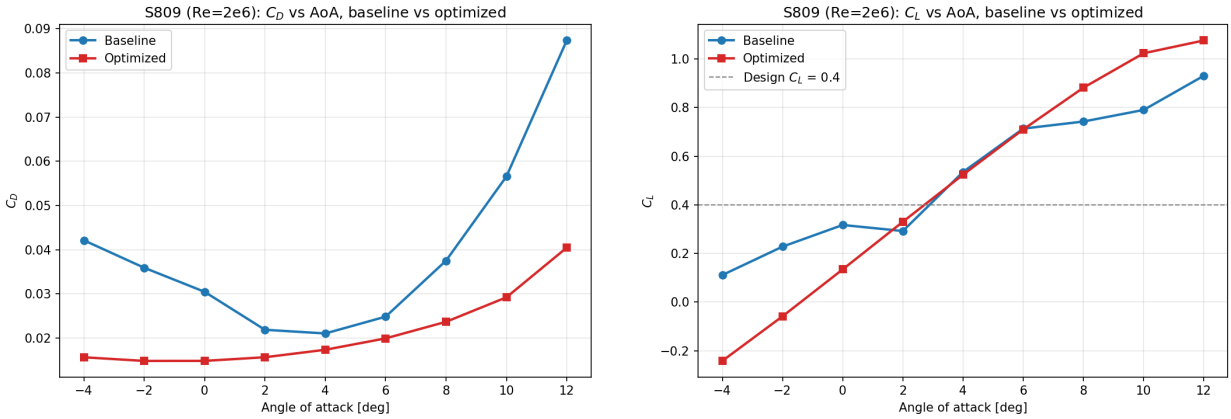


Figure 8: C_D (left) and C_L (right) vs. angle of attack, baseline vs. optimized shape.

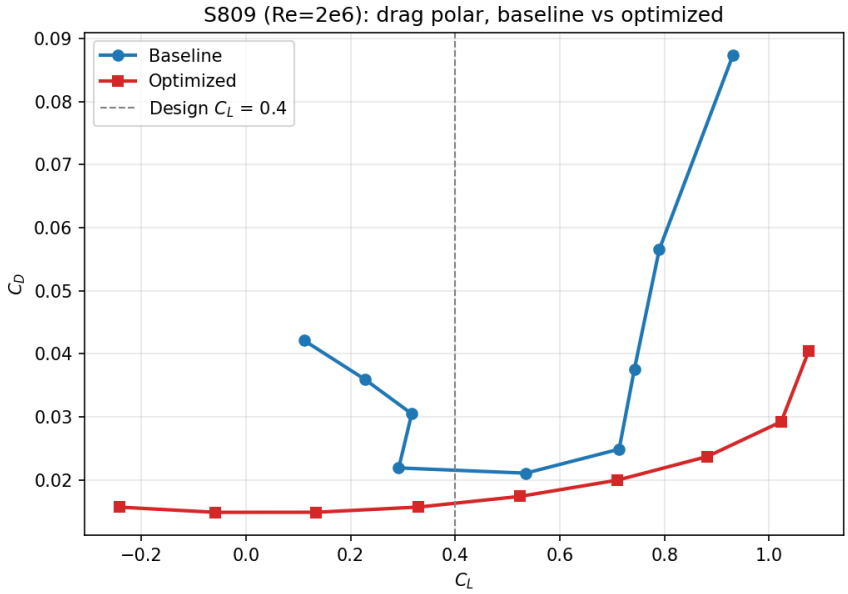


Figure 9: Drag polar (C_D vs. C_L), baseline vs. optimized shape.

The optimized shape shows substantially lower drag than the baseline across nearly the entire angle-of-attack range, with the two C_L–α curves crossing near the design condition ($C_L \approx 0.4$), consistent with the optimizer having shifted the zero-lift angle of attack while preserving performance at the target condition. A small non-monotonic dip is visible in the baseline C_L curve between $\alpha = 0°$ and $2°$ (Figure 8, right); given its small magnitude and that it occurs at only

one of the nine sampled points, this is most likely a slightly under-converged primal solve at that specific angle (the sweep uses a shared warm-started solution across points, without a dedicated convergence-tolerance trim step at each one) rather than a real aerodynamic feature.

7 Baseline vs. Optimized Shape and Pressure Distribution (Case B)

Figure 10 overlays the baseline and optimized airfoil shapes and their surface pressure distributions, both trimmed to $C_L = 0.4$ on the 10,270-cell mesh, extracted directly from the solved CFD surface mesh.

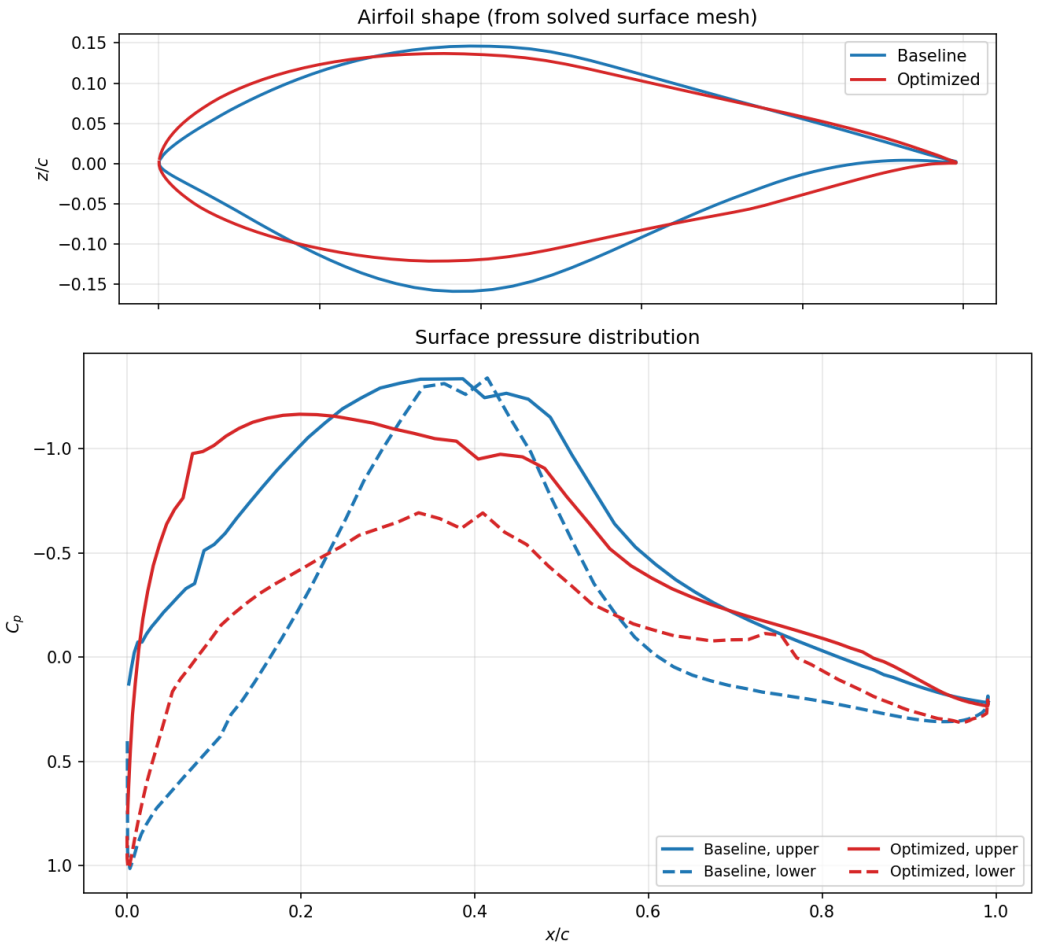


Figure 10: Baseline vs. optimized shape (top) and surface pressure coefficient (bottom), both trimmed to $C_L = 0.4$.

The optimized shape has a fuller, rounder leading edge and reduced maximum camber/thickness on the lower surface relative to the baseline. The pressure distribution shows the mechanism behind

the drag reduction clearly: the optimized shape's upper-surface suction peak is substantially flatter and lower in magnitude ($C_p \approx -0.85$ near $x/c \approx 0.35$) than the baseline's sharper peak ($C_p \approx -1.3$ at a similar location). Spreading the aerodynamic loading more evenly over the chord instead of concentrating it near the leading edge is a well-known drag-reduction mechanism, since it reduces the peak local velocity and the associated viscous losses – fully consistent with the 19–22% drag reduction observed across the mesh convergence study.

8 Discussion

8.1 Case A vs. Case B: Not a Controlled Comparison

It is tempting to compare the 2.79% (Case A) and 22.2% (Case B) drag reductions directly and conclude that CST outperforms FFD, or that optimizing at the higher Reynolds number yields more benefit. This comparison would not be valid: the two cases differ simultaneously in Reynolds number (5×10^5 vs. 2×10^6), mesh resolution (26,522 vs. 10,270 cells), wall treatment (fully wall-resolved vs. wall function), and geometry parametrization (FFD vs. CST). Any of these four differences alone could plausibly account for part of the gap; with all four varying at once, the comparison isolates none of them. A controlled comparison would require running the FFD case at $Re = 2 \times 10^6$ (the case is scaffolded for this in the project but has not yet been executed) so that only the parametrization choice differs.

8.2 Constraint-Limited Optimization

In both cases, the volume constraint (and, in Case A, the leading-edge radius constraint) is active at its bound at the optimum. This indicates that further drag reduction is being limited by the imposed geometric constraints, not by the underlying flow physics – relaxing the volume constraint would be the natural next lever to explore in either case, if a more aggressive design were desired (at the cost of a thinner, potentially less structurally practical section).

8.3 Data Quality Notes

Two minor data-quality points are documented transparently rather than smoothed over: the small non-monotonic dip in the baseline C_L–α curve near $\alpha = 0$–$2°$ (Section 6), and the 80k-mesh baseline point's lift-trim search failing to converge within a reasonable runtime (Section 5), which is why the baseline C_p-vs-mesh-resolution plot has three points instead of four. Neither affects the core drag-reduction results, which are independently verified across multiple mesh resolutions.

9 Conclusions

Two independent adjoint-based CFD shape optimizations of the NREL S809 airfoil were carried out using DAFoam, both minimizing drag coefficient at a fixed target lift coefficient of $C_L = 0.4$:

- **Case A** ($Re = 5 \times 10^5$, FFD, 19 design variables) reduced drag by **2.79%**, converging in 24 IPOPT iterations with the volume and leading-edge-radius constraints active at the optimum.
- **Case B** ($Re = 2 \times 10^6$, CST, 13 design variables) reduced drag by **22.2%**, converging in 12 IPOPT iterations with the volume constraint active at the optimum. This result was confirmed mesh-independent across four mesh resolutions (10k–80k cells, drag reduction settling smoothly from 22.2% to 19.5%), and the underlying mechanism – a flattened, more evenly distributed upper-surface suction peak – was directly verified by comparing the baseline and optimized surface pressure distributions.

The two results are not directly comparable to one another, since Reynolds number, mesh resolution, wall treatment, and geometry parametrization all differ simultaneously between the cases. A natural extension of this work is to run the FFD case at $Re = 2 \times 10^6$, isolating the parametrization choice as the sole variable between an otherwise matched pair of optimizations.

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A Case Directory Reference

Case	Description
`S809_Airfoil/`	Case A: FFD optimization, $Re = 5 \times 10^5$
`airfoil_opt_s809_re2e6_cl04_0000/`	Case B: CST optimization, $Re = 2 \times 10^6$
`airfoil_opt_s809_re2e6_cl04_meshcheck_20k/`	Case B mesh check, 20,792 cells
`airfoil_opt_s809_re2e6_cl04_meshcheck_40k/`	Case B mesh check, 41,340 cells
`airfoil_opt_s809_re2e6_cl04_meshcheck_80k/`	Case B mesh check, 81,984 cells
`airfoil_opt_s809_re2e6_cl04_aoa_sweep_0000/`	Case B angle-of-attack sweep

Table 8: Source case directories for the results presented in this report.